

MATH 3191-001 Fall 2026 Exam 1

23 September 2026

Instructions: Write your name on the paper and show all work that leads to your answers. Answers without work will not be given credit, so you **MUST** justify your answer. You have 1 hour and 15 minutes to take the exam, and there is extra scratch paper available at the front of the room. If you need to use it, please label which problem you are working on and turn it in with the exam, so I can look at your work if necessary.

This exam has 115 points available, but will be scored out of 105, so a score of $\frac{115}{105} \approx 109.5$ is possible.

Question:	1	2	3	4	5	6	7	Total
Points:	20	15	15	20	15	20	10	115
Score:								

Short Answer Questions

1. (20 points total) This problem is walking you through a proof of Invertible Matrix Theorem statement 3 implying statement 6.

- (a) (10 points) Let $A \in \mathbb{R}^{n \times n}$ be row equivalent to an upper triangular matrix B . IE there exist elementary matrices $E_1, \dots, E_k$ such that $E_k \cdots E_1 \cdot A = B$. Argue why if B is invertible, then so is A .

Hint: If you're stuck, think about the determinant of the product of matrices and if the determinant of each elementary matrix can be 0.

Solution: We recall that an elementary matrix is a matrix that is 1 row operation away from the identity. This means that for $\ell = 1, \dots, k$, the diagonal of E_ℓ is non-zero and is upper and/or lower triangular. Thus, for $\ell = 1, \dots, k$, we know that $\det(E_\ell) \neq 0$. From the properties of the determinant, we know that

$$\begin{aligned}\det(E_k \cdots E_1 \cdot A) &= \det(B) \\ \det(E_k) \cdots \det(E_1) \cdot \det(A) &= \det(B) \\ \det(A) &= \frac{\det(B)}{\det(E_k) \cdots \det(E_1)}\end{aligned}$$

If B is invertible, we know that $\det(B) \neq 0$, so from above we know $\det(A) \neq 0$, which lets us conclude that A is invertible as well.

- (b) (10 points) Prove that if an upper triangular matrix $A \in \mathbb{R}^{n \times n}$ has a pivot in every column, then show the columns of A are linearly independent

Hint: Recall what having n pivots means about the number of solutions to a given system!

Solution: Recall that for a matrix A to have n pivots then the diagonal elements of it must be non-zero. This means the augmented matrix $[A \mid \mathbf{0}]$ can be row reduced to $[I \mid \mathbf{0}]$. Thus, the homogeneous system has only the trivial solution. In other words if we consider A as $A = [\mathbf{a}_1 \ \dots \ \mathbf{a}_n]$ then the system $c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n$ has only the solution $(c_1, \dots, c_n) = (0, \dots, 0)$ which, by the definition of linear independence means that $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly independent.

Note. Combining part (a) and (b) would let us say that any matrix $A \in \mathbb{R}^{n \times n}$ being row equivalent to an upper triangular matrix with n pivots has linearly independent columns.

2. (15 points) Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, describe what information you need to determine if T is invertible

Hint: think about the invertible matrix theorem and a condition on m and n .

Solution: 5 points each for (1) requiring $m = n$, (2) alluding to the invertible matrix theorem, and (3) how T acts on n linearly independent vectors (will give credit if linearly independent is not explicitly stated)

Sample solution:

First, we need to know that $m = n$, and then we need to know how T acts on n linearly independent vectors. Then with this information, we can construct the matrix associated with this linear transformation and check if A is row-equivalent to an upper triangular matrix with n pivots. If all of this is true, then we get to say that T is an invertible linear transformation. Otherwise, we would say that T is not an invertible linear transformation.

Computational Question

3. (15 points) Let $A \in \mathbb{R}^{2 \times 3}$, $\mathbf{b} \in \mathbb{R}^2$ be as given below. Determine the solution set that solves the equation $A\mathbf{x} = \mathbf{b}$.

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 7 & 30 & 12 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 6 \\ 44 \end{bmatrix}$$

Hint: your final answer should contain only integers

Solution: We will construct the following augmented matrix

$$\left[A \mid \mathbf{b} \right] = \left[\begin{array}{ccc|c} 1 & 4 & 2 & 6 \\ 7 & 30 & 12 & 44 \end{array} \right]$$

Which we row reduce as follows

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 6 \\ 7 & 30 & 12 & 44 \end{array} \right] & \xrightarrow{R_2 \leftarrow R_2 - 7R_1} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 6 \\ 0 & 2 & -2 & 2 \end{array} \right] \\ & \xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 6 \\ 0 & 1 & -1 & 1 \end{array} \right] \\ & \xrightarrow{R_1 \leftarrow R_1 - 4R_2} \left[\begin{array}{ccc|c} 1 & 0 & 6 & 2 \\ 0 & 1 & -1 & 1 \end{array} \right] \end{aligned}$$

This lets us construct our solution set as

$$\left\{ \mathbf{x} \in \mathbb{R}^3 \mid \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -6 \\ 1 \\ 1 \end{bmatrix} \text{ for some } t \in \mathbb{R} \right\}$$

4. (20 points) Let $A \in \mathbb{R}^{3 \times 2}$, $\mathbf{b} \in \mathbb{R}^3$ be as given below, determine a value $z \in \mathbb{R}$ such that $A\mathbf{x} = \mathbf{b}$ is consistent (IE: has at least 1 solution)

$$A = \begin{bmatrix} 2 & z \\ 4 & 4 \\ 6 & 2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix}$$

Solution: We will first set up the augmented system:

$$\left[A \mid \mathbf{b} \right] = \left[\begin{array}{cc|c} 2 & z & 1 \\ 4 & 4 & 4 \\ 6 & 2 & 4 \end{array} \right]$$

which we row reduce as follows

$$\left[\begin{array}{cc|c} 2 & z & 1 \\ 4 & 4 & 4 \\ 6 & 2 & 4 \end{array} \right] \xrightarrow[\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - 3R_1}]{\substack{R_3 \leftarrow R_3 - 3R_1 \\ R_2 \leftarrow R_2 - 2R_1}} \left[\begin{array}{cc|c} 2 & z & 1 \\ 0 & 4 - 2z & 2 \\ 0 & 2 - 3z & 1 \end{array} \right]$$

Now, we want this system to be consistent. Since we only have two variables, we must be able to get the final row to be a 0 row. Looking at the last column, the only way this can happen is if $R_3 \leftarrow R_3 - 2R_2$ results in the entire row being 0. In other words: we need z to satisfy:

$$4 - 2z = 4 - 6z \rightarrow 4z = 0$$

This means that z must be 0.

There are other methods that we can do as well!

5. (15 points) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation as given below. Determine the matrix associated A associated with T ie: compute A such that $A\mathbf{x} = T(\mathbf{x})$.

$$T\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix} \quad T\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 7 \\ 1 \\ 3 \end{bmatrix}$$

Solution: We recall that the matrix associated with a linear transformation is defined to be

$$A = [T(\mathbf{e}_1) \quad \dots \quad T(\mathbf{e}_n)].$$

This means we need to be able to write $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ as a linear combination of $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$. We will do this as we discussed on Monday in class; by row reducing the augmented matrix given by

$$[\mathbf{v}_1 \quad \mathbf{v}_2 \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array} \right].$$

Which we do as follows:

$$\begin{aligned} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array} \right] & \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array} \right] \\ & \xrightarrow{R_2 \leftarrow -R_2} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array} \right] \\ & \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & 1 & 3 & -1 \end{array} \right] \end{aligned}$$

This means that we can write $\mathbf{e}_1 = -5\mathbf{v}_1 + 3\mathbf{v}_2$ and $\mathbf{e}_2 = 2\mathbf{v}_1 - \mathbf{v}_2$. Thus we get

$$\begin{aligned} T(\mathbf{e}_1) &= -5T(\mathbf{v}_1) + 3T(\mathbf{v}_2) \\ &= -5 \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} 7 \\ 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 16 \\ -27 \\ -6 \end{bmatrix} \\ T(\mathbf{e}_2) &= 2T(\mathbf{v}_1) - T(\mathbf{v}_2) \\ &= 2 \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix} - \begin{bmatrix} 7 \\ 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} -5 \\ 11 \\ 3 \end{bmatrix} \end{aligned}$$

Finally, we get that

$$A = \begin{bmatrix} 16 & -5 \\ -27 & 11 \\ -6 & 3 \end{bmatrix}$$

Hint: Your final answer will have some large numbers, but you should not have to row reduce a matrix that looks too disgusting.

Challenge Question

6. (20 points) Let k, n be natural numbers¹ such that $k > n$. Let $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$. Argue why this list must be linearly dependent.

Solution: If we construct the augmented matrix

$$\left[\begin{array}{ccc|c} \mathbf{v}_1 & \dots & \mathbf{v}_k & \mathbf{0} \end{array} \right]$$

then we have a $n \times k + 1$ matrix. Since the final column is a column of 0s we will never get a pivot in this final column. We note that can have at most $n < k$ pivots as we can only have 1 per row. Thus, we will have at least $k - n > 0$ free variables. Therefore, the system

$$a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k = \mathbf{0}$$

will have infinitely many solutions, which means $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly dependent as desired.

¹The natural numbers are integers greater than 0, in other words: $\{1, 2, \dots\}$

Bonus Question

7. (10 points) Answer one of the following questions for 10 points. You may answer both, but I will only give you points for one!
1. Define the slang term “glow up” particularly in the context of: “This person had a glow up.”
 2. Draw a picture of your favorite animal or character: either original or in media